

Assignment 3: Unsupervised Learning

Assigned: 4.16

Due: 5.7 23:59 p.m.

1 Written Problems (50 points)

1.1. Mixture of Bernoulli In lecture we introduced the Gaussian mixture model (GMM). In this question, however, we consider a mixture of Bernoulli distributions. In this case, our datapoints are denoted as m -dimensional vectors of binary values $\mathbf{x} \in \{0, 1\}^m$.

Let $\boldsymbol{\mu}_k$ denotes a mean vector, a single multivariate Bernoulli distribution can be formulated as

$$p(\mathbf{x}|\boldsymbol{\mu}_k) = \prod_{j=1}^m \mu_k[j]^{\mathbf{x}[j]} (1 - \mu_k[j])^{1-\mathbf{x}[j]},$$

where $\mu_k[j]$ and $\mathbf{x}[j]$ denotes the j -th value of $\boldsymbol{\mu}_k$ and $\mathbf{x}$, respectively. And we define a parameter set of mixture as $\Theta = \{\boldsymbol{\mu}_k, \pi_k | k = 1, \dots, K\}$

- (3 points) write a mixture of K multivariate Bernoulli distributions $p(\mathbf{x}|\Theta)$.
- (4 points) suppose we have dataset $X = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ (this definition also works in the following problem), write an expression for log-likelihood $L(\Theta)$
- (6 points) Now, suppose that we divide the datapoints $\mathbf{x}$ into two parts $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\mathbf{x}_1 \in \{0, 1\}^{m-d}$ and $\mathbf{x}_2 \in \{0, 1\}^d$. Please show that the conditional distribution $p(\mathbf{x}_1|\mathbf{x}_2; \Theta)$ is also a Bernoulli mixture distribution and provide expressions for the mixing coefficients and the component densities of this new mixture.

In the following question, we first introduce a binary latent variable $\mathbf{z}^{(i)}$ associated with each instance of $\mathbf{x}^{(i)}$. Unlike Gaussian mixture, here $\mathbf{z} = (z_1, \dots, z_K)^\top$ is a binary K -dimensional variable having a single component equal to 1, with all other components equal to 0. Let $Z = \{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(n)}\}$

- (6 points) What is $p(\mathbf{z}^{(i)}|\pi)$, $p(\mathbf{x}^{(i)}|\mathbf{z}^{(i)}, \Theta)$, and $p(Z, X|\Theta)$?
- (6 points) Let $\phi(z_k^{(i)}) = \mathbb{E}[z_k^{(i)}|\mathbf{x}^{(i)}, \Theta]$. Show that

$$\phi(z_k^{(i)}) = \frac{\pi_k \prod_{j=1}^m \mu_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \mu_k[j])^{1-\mathbf{x}^{(i)}[j]}}{\sum_k \pi_k \prod_{j=1}^m \mu_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \mu_k[j])^{1-\mathbf{x}^{(i)}[j]}}$$

- (4 points) Derive the E-step expression $\mathbb{E}[\ln p(Z, X|\Theta)|X, \Theta]$
- (6 points) Show that the values of $\tilde{\boldsymbol{\mu}}_k$ and $\tilde{\pi}_k^*$ that maximize the E-step expression are

$$\tilde{\boldsymbol{\mu}}_k = \frac{\sum_{i=1}^n \phi(z_k^{(i)}) \mathbf{x}^{(i)}}{N_k}, \quad \tilde{\pi}_k = \frac{N_k}{\sum_k N_k}$$

where $N_k := \sum_{i=1}^n \phi(z_k^{(i)})$. Remark: consider the constraint $\sum_{k=1}^K \pi_k - 1 = 0$

Solution:

- $p(\mathbf{x}|\Theta) = \sum_{k=1}^K \pi_k p(\mathbf{x}|\boldsymbol{\mu}_k) = \sum_{k=1}^K \pi_k \prod_{j=1}^m \boldsymbol{\mu}_k[j]^{\mathbf{x}[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{x}[j]}$.
- $L(\Theta) = \sum_{i=1}^n \ln p(\mathbf{x}^{(i)}|\Theta) = \sum_{i=1}^n \ln \sum_{k=1}^K \pi_k \prod_{j=1}^m \boldsymbol{\mu}_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{x}^{(i)}[j]}$.
- First, we can apply the rules of probability to find that

$$p(\mathbf{x}_1|\mathbf{x}_2;\Theta) = \frac{p(\mathbf{x}_1, \mathbf{x}_2|\Theta)}{p(\mathbf{x}_2|\Theta)} = \frac{p(\mathbf{x}|\Theta)}{p(\mathbf{x}_2|\Theta)} = \frac{\sum_{k=1}^K \pi_k p(\mathbf{x}|\boldsymbol{\mu}_k)}{p(\mathbf{x}_2|\Theta)} = \sum_{k=1}^K \frac{\pi_k}{p(\mathbf{x}_2|\Theta)} p(\mathbf{x}|\boldsymbol{\mu}_k)$$

Thus, we see that $p(\mathbf{x}_1|\mathbf{x}_2;\Theta)$ is a mixture distribution with the same component density as before but with the mixture coefficients given by $\frac{\pi_k}{p(\mathbf{x}_2|\Theta)}$. The term $p(\mathbf{x}_2|\Theta)$ can be computed via marginalization as

$$p(\mathbf{x}_2|\Theta) = \sum_{\mathbf{c} \in \{0,1\}^{m-d}} \sum_{k=1}^K \pi_k \left(\prod_{j=1}^{m-d} \boldsymbol{\mu}_k[j]^{\mathbf{c}[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{c}[j]} \right) \left(\prod_{j=m-d+1}^m \boldsymbol{\mu}_k[j]^{\mathbf{x}_2[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{x}_2[j]} \right)$$

(This term is not necessary when grading)

- $p(\mathbf{z}^{(i)}|\pi) = p(z_1^{(i)} \cdots z_K^{(i)}|\pi) = \prod_{k=1}^K (\pi_k)^{z_k^{(i)}}$
 $p(\mathbf{x}^{(i)}|\mathbf{z}^{(i)}, \Theta) = \prod_{k=1}^K \left(p(\mathbf{x}^{(i)}|\mathbf{z}^{(i)}, \Theta, A_k^{(i)}) \right)^{z_k^{(i)}} = \prod_{k=1}^K \left(p(\mathbf{x}^{(i)}|\boldsymbol{\mu}_k) \right)^{z_k^{(i)}}$

$$\begin{aligned} p(Z, X|\Theta) &= \prod_{i=1}^n p(\mathbf{z}^{(i)}, \mathbf{x}^{(i)}|\Theta) = \prod_{i=1}^n p(\mathbf{x}^{(i)}|\mathbf{z}^{(i)}, \Theta) p(\mathbf{z}^{(i)}|\pi) \\ &= \sum_{i=1}^n \left(\prod_{k=1}^K (\pi_k)^{z_k^{(i)}} \right) \left(\prod_{k=1}^K \left(p(\mathbf{x}^{(i)}|\boldsymbol{\mu}_k) \right)^{z_k^{(i)}} \right) \end{aligned}$$

•

$$\begin{aligned} \phi(z_k^{(i)}) &= \mathbb{E} \left[z_k^{(i)} | \mathbf{x}^{(i)}, \Theta \right] \\ &= p(z_k^{(i)} = 1 | \mathbf{x}^{(i)}, \Theta) \\ &= \frac{p(z_k^{(i)} = 1, \Theta) p(\mathbf{x}^{(i)} | z_k^{(i)} = 1, \Theta)}{\sum_k p(z_k^{(i)} = 1, \Theta) p(\mathbf{x}^{(i)} | z_k^{(i)} = 1, \Theta)} \\ &= \frac{\pi_k \prod_{j=1}^m \boldsymbol{\mu}_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{x}^{(i)}[j]}}{\sum_k \pi_k \prod_{j=1}^m \boldsymbol{\mu}_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \boldsymbol{\mu}_k[j])^{1-\mathbf{x}^{(i)}[j]}} \quad \square \end{aligned}$$

•

$$\begin{aligned} \ln P(Z, X|\Theta) &= \sum_{i=1}^n \left(\left(\sum_{k=1}^K z_k^{(i)} \ln \pi_k \right) + \left(\sum_{k=1}^K \left(z_k^{(i)} \ln p(\mathbf{x}^{(i)}|\boldsymbol{\mu}_k) \right) \right) \right) \\ &= \sum_{i=1}^n \sum_{k=1}^K z_k^{(i)} \left(\ln \pi_k + \ln p(\mathbf{x}^{(i)}|\boldsymbol{\mu}_k) \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^n \sum_{k=1}^K z_k^{(i)} \left(\ln \pi_k + \ln \prod_{j=1}^m \mu_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \mu_k[j])^{1 - \mathbf{x}^{(i)}[j]} \right) \\
&= \sum_{i=1}^n \sum_{k=1}^K z_k^{(i)} \left(\ln \pi_k + \sum_{j=1}^m \mathbf{x}^{(i)}[j] \ln \mu_k[j] + (1 - \mathbf{x}^{(i)}[j]) \ln(1 - \mu_k[j]) \right)
\end{aligned}$$

Let $\tilde{\mu}, \tilde{\pi}$ be the new parameters that we'd like to maximize, and taking the expected value:

$$\mathbb{E}[\ln P(Z, X|\Theta)|X, \Theta] = \sum_{i=1}^n \sum_{k=1}^K \phi(z_k^{(i)}) \left(\ln \tilde{\pi}_k + \sum_{j=1}^m \mathbf{x}^{(i)}[j] \ln \tilde{\mu}_k[j] + (1 - \mathbf{x}^{(i)}[j]) \ln(1 - \tilde{\mu}_k[j]) \right)$$

(There are quite a lot of notations here. If students can give the answer of $\ln P(Z, X|\Theta)$, they can get full marks.)

- Setting the derivative to 0:

$$\frac{\partial \mathbb{E}[\ln P(Z, X|\Theta)|X, \Theta]}{\partial \tilde{\mu}_k[j]} = \sum_{i=1}^n \phi(z_k^{(i)}) \left(\frac{\mathbf{x}^{(i)}[j]}{\tilde{\mu}_k[j]} - \frac{1 - \mathbf{x}^{(i)}[j]}{1 - \tilde{\mu}_k[j]} \right) = 0$$

Multiply by the denominators:

$$\begin{aligned}
&\sum_{i=1}^n \phi(z_k^{(i)}) \left((1 - \tilde{\mu}_k[j]) \mathbf{x}^{(i)}[j] - \tilde{\mu}_k[j] (1 - \mathbf{x}^{(i)}[j]) \right) = \sum_{i=1}^n \phi(z_k^{(i)}) \left(\mathbf{x}^{(i)}[j] - \tilde{\mu}_k[j] \right) \\
&\Rightarrow \tilde{\mu}_k[j] = \frac{\sum_{i=1}^n \phi(z_k^{(i)}) \mathbf{x}^{(i)}[j]}{N_k} \\
&\Rightarrow \tilde{\mu}_k = \frac{\sum_{i=1}^n \phi(z_k^{(i)}) \mathbf{x}^{(i)}}{N_k} \quad \square
\end{aligned}$$

For $\tilde{\pi}$, we only need to minimize $\sum_{i=1}^n \sum_{k=1}^K \phi(z_k^{(i)}) \ln \tilde{\pi}$ since the rest is not a function of $\tilde{\pi}$. In order to keep $\tilde{\pi}$ a distribution, we require $\sum_{k=1}^K \tilde{\pi} = 1$. Let λ be the dual variable for this constraint:

$$L(\pi, \lambda) = - \sum_{i=1}^n \sum_{k=1}^K \phi(z_k^{(i)}) \ln \pi_k + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

Taking the derivate w.r.t. π_k :

$$\frac{\partial L(\pi, \lambda)}{\partial \pi_k} = \lambda - \sum_{i=1}^n \frac{\phi(z_k^{(i)})}{\pi_k} = 0$$

Now we solve for λ :

$$L(\pi) = - \sum_{i=1}^n \sum_{k=1}^K \phi(z_k^{(i)}) (\ln N_k - \ln \lambda) + \left(\sum_{k=1}^K N_k - \lambda \right)$$

Take the derivative w.r.t. λ :

$$\lambda = \sum_{k=1}^K N_k$$

Hence we have

$$\pi_k = \frac{N_k}{\sum_{k=1}^K N_k} \quad \square$$

1.2. Confusion Matrix

		PREDICTED		
		Classes		
		Positive (1)	Negative (0)	Total
ACTUAL	Positive (1)	TP = 20	FN = 5	25
	Negative (0)	FP = 10	TN = 15	25
		Total		
		30	20	50

Figure 1: Two-class confusion matrix

- (4 points) See a case of two-class confusion matrix shown in Figure 1, calculate the accuracy, precision, recall, and F1-Score with the definition:

$$\text{F1score} = \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}}$$

- (2 points) Classification tasks in machine learning involving more than two classes are known by the name of "multi-class classification". See a case of multi-class confusion matrix shown in Figure 2. What are the TP, FP, FN, TN for class b ?
- (9 points) Calculate the accuracy, macro average precision, macro average recall, and balanced accuracy of Figure 2 with the definition:

$$\begin{aligned} \text{MacroPrecision} &= \frac{\sum_{k=1}^K TP_k / (TP_k + FP_k)}{K} \\ \text{MacroRecall} &= \frac{\sum_{k=1}^K TP_k / (TP_k + FN_k)}{K} \\ \text{BalancedAccuracy} &= \frac{\sum_{k=1}^K TP_k / (Total_{row_k})}{K} \end{aligned}$$

where k denotes the class label. In this example, is the balanced accuracy the same as the accuracy? Why?

Solution:

		PREDICTED classification					
		<i>Classes</i>	a	b	c	d	Total
ACTUAL classification	a	5	23	17	17		62
	b	10	540	21	14		585
	c	166	96	436	110		808
	d	1	2	5	87		95
	Total	182	661	479	228		1550

Figure 2: Multi-class Confusion Matrix

		PREDICTED classification				
		Classes	a	b	c	d
ACTUAL classification	a	TN	FP	TN	TN	
	b	FN	TP	FN	FN	
	c	TN	FP	TN	TN	
	d	TN	FP	TN	TN	

$$\begin{aligned}
 accuracy &= 0.7, \\
 precision &= 0.66, \\
 recall &= 0.8, \\
 f1score &= 0.72
 \end{aligned}$$

$$\begin{aligned}
 accuracy &= 0.69, \\
 marcoprecision &= 0.53, \\
 marcorecall &= 0.61, \\
 balancedaccuracy &= 0.615
 \end{aligned}$$

Accuracy value is 0.689, whereas Balanced Accuracy is 0.615. The difference is mainly due to the weighting that recall applies on each row/actual class. In this way each class has an equal weight in the final calculation of Balanced Accuracy and each class is represented by its recall, regardless of their size. Accuracy instead, mainly depends on the performance that

the algorithm achieves on the biggest classes. The performance on the smallest ones is less important, because of their low weight.

2 Programming (50 points)

Important Notes

1. All instructions and questions are provided in the accompanying Jupyter notebook file **Assignment3_KMeans_GMM.ipynb**.
2. Please make sure to complete all the code and analysis cells. Specifically, fill in all sections marked with comments such as `## Write your code here ##`.
3. You are not required to copy your answers from the notebook into the written report. However, you must submit the completed notebook file as part of your assignment.
4. Do not include any datasets or generated data files in your submission.
5. Ensure that all code cells are executed and outputs (such as plots, printed results, and metrics) are visible in your final notebook. Incomplete or non-executed notebooks will receive point deductions.

2.1. (K-Means Clustering, 25 points) In this section, you will implement the K-Means clustering algorithm from scratch. This includes initializing cluster centers, assigning points to clusters based on distance, updating centers iteratively, and evaluating clustering quality using the Silhouette Score. You will also visualize the clustering results and interpret the output.

2.2. (Gaussian Mixture Model with EM, 25 points) In this section, you will implement a Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) algorithm. This involves computing the multivariate Gaussian density, implementing both the E-step and M-step, and fitting the model to the dataset. You will visualize the resulting clusters and evaluate model performance using the Silhouette Score.